Two evaporation regimes for emergent-gravity black holes: a machine-checked partition at the induced-gravity cutoff

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(Dated: November 7, 2026)

A black hole on a substrate whose Newton constant is induced by fermion loops below an ultraviolet cutoff has two evaporation regimes, told apart by the sign of $dT_{\rm H}/dt$. Above a critical mass it heats as it evaporates; below it, it cools toward extremality in infinite time. That critical mass is the one whose emergent gravitational length reaches the cutoff length. We formalize the partition in Lean 4: a decidable classifier, strict monotonicity on each branch, a criterion theorem fixing the sign of $dT_{\rm H}/dt$ from a mass comparison, and falsifier theorems stating what would refute it.

INTRODUCTION

A Schwarzschild black hole has negative heat capacity: it grows hotter as it radiates. Its temperature $T_{\rm H} = \hbar/(8\pi M)$ rises without bound as the mass falls, and a hole of initial mass M_0 disappears after a time of order M_0^3 [1]. A sonic horizon in a flowing fluid does the opposite. Balbinot, Fagnocchi and Fabbri solved the linearized backreaction of the emitted phonons on the flow and found that the horizon migrates, the supersonic region shrinks, and the emission temperature falls linearly in time at leading order, with a tail $t \sim T_{\rm H}^{-3}$ that sends $T_{\rm H}$ to zero only after infinite time [2]. They compare that end state to a near-extremal Reissner–Nordström hole, and we use *extremality* throughout in that sense: a vanishing temperature that is approached and never attained.

Those are opposite signs of the same derivative. Each is derived on its own background, and on a fixed background there is nothing for a criterion to decide. The question becomes sharp on a substrate where the gravitational coupling is itself induced by the matter it couples to. One system then carries a geometric description at large mass and a fluid description at small mass, and something has to separate them.

This Letter supplies that separation and machine-checks it. On a substrate whose Newton constant is induced by N_f Dirac species below a cutoff $\Lambda_{\rm UV}$, the sign of $dT_{\rm H}/dt$ under evaporation changes at a single mass,

$$M_{\rm c} = \frac{N_f\,\Lambda_{\rm UV}}{12\pi\,\alpha_{\rm ADW}}, \tag{1}$$

where $\alpha_{\rm ADW}$ measures the departure of the induced coupling from its bare one-loop value. The scale is fixed once the substrate is, and it is not a dimensional guess: Eq. (1) says exactly that the emergent gravitational length of $M_{\rm c}$ equals the shortest length the substrate resolves.

The formal development comprises 16 theorems across 28 declarations, in Lean 4.32.0 against Mathlib revision 81a5d257. The kernel accepts them from `propext`, `Classical.choice` and `Quot.sound` alone, with no substrate-specific axiom and no incomplete proof. Which hypotheses the partition rests on is stated where each is used, and Sec. gives the falsifier theorems whose content is what would refute the construction.

THE CRITICAL MASS

Sakharov observed that the one-loop effective action of matter fields in a curved background already contains an Einstein–Hilbert term, so Newton's constant need not be fundamental [3, 4]. For N_f Dirac species regulated by a hard cutoff the induced coupling is $G_N^{\rm Sak} = 12\pi/(N_f\Lambda_{\rm UV}^2)$. We write $G_N^{\rm emerg} = \alpha_{\rm ADW}\,G_N^{\rm Sak}$ for a substrate whose gravitational response departs from that value, with $\alpha_{\rm ADW} = 1$ the bare induced limit. The numerical coefficient is regulator dependent; what survives regulator choice is that the induced coupling is finite and of definite sign [5].

Substituting $G_N^{\rm Sak}$ into Eq. (1) gives an identity:

$$G_N^{\rm emerg}\,M_{\rm c} \;=\; \Lambda_{\rm UV}^{-1}. \tag{2}$$

$M_{\rm c}$ is the mass whose emergent gravitational length is the shortest length the substrate resolves. A hole heavier than $M_{\rm c}$ has a horizon large compared with the cutoff and admits a geometric description. A hole lighter than $M_{\rm c}$ would need a horizon smaller than the substrate's own resolution, and what responds is the fluid rather than the geometry. Read this way, Eq. (1) is the mass at which the two descriptions exchange roles rather than a fitted scale. Equation (2) is also where the construction's one free choice sits: no derivation on this substrate pins the crossover at $G_N^{\rm emerg}M = \Lambda_{\rm UV}^{-1}$ rather than at an order-unity multiple of it, so the coefficient in Eq. (1) is an ansatz even though its parametric form is not.

A hole is described by its mass, its horizon area and its surface gravity κ, and a substrate by $(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$, each strictly positive; $M_{\rm c}$ is then positive wherever it is defined (`M_c`, `M_c_pos`). Comparing M against $M_{\rm c}$ sorts a hole into one of three regimes, and the sorting is decidable and reversible: each regime label holds if and only if the corresponding mass inequality does (`classify`, `classify_Schwarzschild_iff`,

`classify_ADWExtremality_iff`, `classify_Boundary_iff`). A regime label and a mass comparison therefore carry the same information, and neither can drift from the other.

ONE SUBSTRATE, BOTH BRANCHES

The strongest reason to want a criterion rather than an analogy is that a single material realizes both signs.

A moving domain wall in a thin film of ^{3}He-A obeys $T_{\rm H}(v) = T_{\rm H}^{(0)}(1 - v^2/c_\perp^2)$, with $T_{\rm H}^{(0)} = \hbar c_\perp/(2\pi d)$ fixed by the wall thickness d [7]. The temperature falls with the wall speed v and stays finite as $v \to 0$. Radiation is fed by the kinetic energy of the wall and slows it, so $T_{\rm H}$ rises in time and the end temperature does not vanish. The thin film sits on the heating branch, and Balbinot *et al.* draw exactly that contrast, noting that such a film appears to show a non-vanishing end temperature of the evaporation process [2].

A moving soliton in bulk ^{3}He-A sits on the other branch. Its horizon exists only for $v > c_\perp$, and its surface gravity carries a factor $(1 - c_\perp^2/v^2)^{1/2}$ that vanishes as backreaction slows the soliton toward $c_\perp$; Jacobson and Volovik accordingly describe that system as cooling toward an extremal configuration [6]. Two textures in the same superfluid, evaporating in opposite directions, is what a criterion has to resolve.

THE TWO BRANCHES AND THE CRITERION

Above $M_{\rm c}$ the profile is the textbook one, $T_{\rm H} = 1/(8\pi M)$ in natural units. It is positive and strictly decreasing in M (`T_H_schwarzschild_pos`, `T_H_schwarzschild_strict_decreasing`), so evaporation, which drives $dM/dt < 0$, raises the temperature and the lifetime is finite [1].

Below $M_{\rm c}$ the profile is the sonic one. Writing κ for the surface gravity of the sonic horizon, c for the sound speed, A_0 for the amplitude of the quantum source driving the backreaction and ϵ for the expansion parameter that switches it on, Balbinot *et al.* obtain to leading order

$$T_{\rm H}(t) = \frac{\hbar c}{2\pi}\,\kappa\left[1 - \frac{563}{720\pi}\,\epsilon\,\kappa^3\,c\,A_0\,t\right], \qquad (3)$$

together with the extrapolation $t \sim T_{\rm H}^{-3}$ that sends $T_{\rm H}$ to zero only as $t \to \infty$ [2]. The profile carried through the development is instead the exponential $T_{\rm H}(t) = T_{\rm H}^{(0)} e^{-t/\tau_{\rm cool}}$, positive and strictly decreasing for positive $T_{\rm H}^{(0)}$ and $\tau_{\rm cool}$, and equal to $T_{\rm H}^{(0)}$ at $t = 0$ (`T_H_acoustic_evolution` and its accompanying positivity, monotonicity and initial-value lemmas).

Substituting one for the other is a scope condition on everything proved of the cooling branch, and it is worth stating twice rather than not at all. The exponential agrees with Eq. (3) at $t = 0$ and shares the two features the partition consumes, a temperature that falls throughout the evaporation and reaches zero only in the infinite-time limit, but it is a different function. What is proved is proved of a representative of the cooling branch and not of an exact evaporation history. Equation (3) is itself a leading order and holds for a fluid of constant sound speed, so a condensate, where $c \propto \rho^{1/2}$, inherits the branch by the sign of $dT_{\rm H}/dt$ rather than by that formula.

The partition itself is a single theorem. A substrate satisfying positivity of $M_{\rm c}$ and of the initial temperature, a positive evaporation rate above $M_{\rm c}$, a negative one below, and the substrate response $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$, has an evaporation rate whose sign is fixed by a mass comparison alone: positive when $M > M_{\rm c}$, negative when $M < M_{\rm c}$ (`H_RegimePartition`, `regime_partition_criterion`). The two branch signs enter as hypotheses, each grounded in its own primary source, Ref. [1] above $M_{\rm c}$ and Ref. [2] below it. What the criterion adds is that the comparison against $M_{\rm c}$ decides between them, which is the statement neither source makes.

Two of the four laws of black-hole mechanics enter the development, and neither in full. The first-law slot fixes the positivity $0 < G_N^{\rm emerg}$ that the coupling $1/(8\pi G_N^{\rm emerg})$ requires, together with the response identity above; the differential first law itself is not formalized. The second-law slot records the Glorioso–Liu conclusion, that the entropy-current divergence is non-negative, as a hypothesis rather than as a consequence, because deriving it needs the Schwinger–Keldysh effective action [11–13]. Constancy of κ over the horizon, and any affine-time statement of the third law, are absent for want of Killing-vector and affine-time machinery. What does follow unconditionally is that a hole classified into the cooling branch and satisfying those two slots has a strictly negative evaporation rate (`FourLaws_ADWExtremality`, `four_laws_consistent_with_acoustic_regime`).

WHAT WOULD REFUTE THIS

A construction assembled from hypotheses is worth stating only if the hypotheses can fail, so each load-bearing component is paired with a theorem that derives a contradiction from its denial.

A single hole above $M_{\rm c}$ observed to cool as it evaporates refutes the construction, with no appeal to how the cooling was modelled: the partition hypotheses, a heating-branch classification and a non-positive rate are together contradictory (`falsifier_schwarzschild_heating`).

On the cooling branch the refutable content is a comparison of two measurements rather than a func-

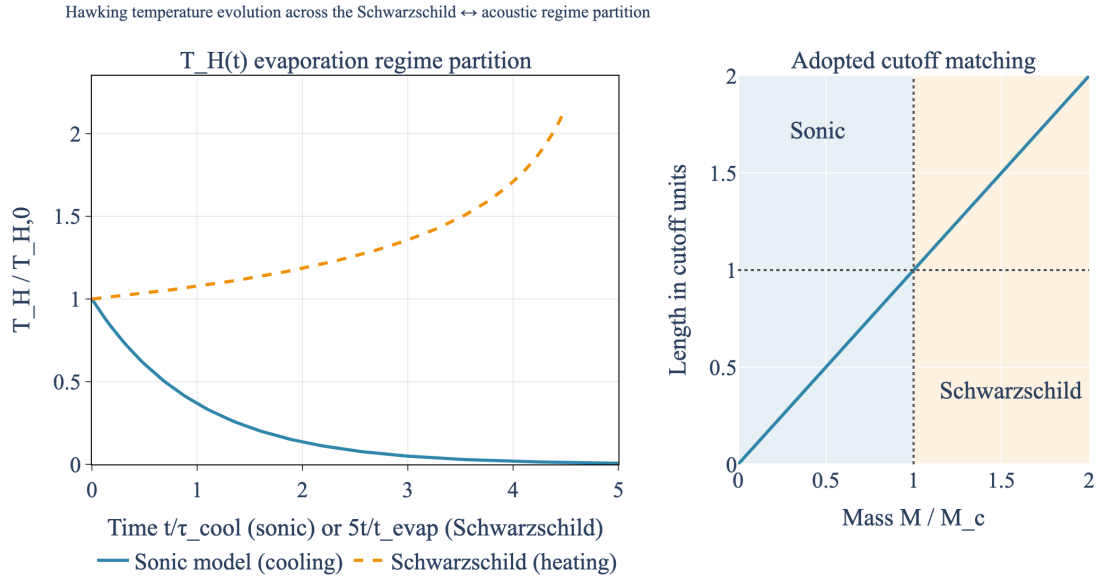


FIG. 1. Left: illustrative evaporation trajectories in units of each branch's initial temperature. The time coordinate is $t/\tau_{\rm cool}$ for the sonic model and $5t/t_{\rm evap}$ for Schwarzschild. Right: the emergent gravitational length $G_N^{\rm emerg}M$ (solid) and the cutoff length $\Lambda_{\rm UV}^{-1}$ (horizontal), both in units of the cutoff length. The adopted matching condition is $M = M_{\rm c}$ [Eqs. (1) and (2)].

tional form. Any alternative history that agrees with the profile at the start of an interval, and fails to decrease across it, must disagree at the end (`falsifier_acoustic_decay_form`).

The third law enters through the time $\tau(T_\star)$ needed to reach $T_{\rm H} = T_\star$. Its strong form requires $\tau(T_\star)$ to exceed any prescribed bound for all small enough $T_\star$ [9]. Kehle and Unger construct regular collapse data for the Einstein–Maxwell-charged scalar field system that reach an exactly extremal horizon at finite advanced time, which bounds τ uniformly on such a tail [10]. The two are contradictory (`falsifier_third_law_form`), so a substrate must choose and the third-law slot cannot be filled vacuously. The theorem selects neither alternative.

The substrate response is constrained from both sides. It is strictly positive whenever $\alpha_{\rm ADW} > 1$, and nonzero exactly when $\alpha_{\rm ADW} \neq 1$ (`falsifier_alpha_ADW_dependence`, `delta_ADW_nonzero_iff_alpha_ADW_ne_one`). A substrate with $\alpha_{\rm ADW} > 1$ and a non-positive response is excluded, and so is one measured away from the bare induced limit with a vanishing response.

DISCUSSION

The result has two limits worth stating. No closed-form $T_{\rm H}(M)$ interpolating the regimes is derived: $M = M_{\rm c}$ is the value at which the classifier changes branch, not a demonstrated smooth crossover, and no single analytic profile is shown to join the branches there. The entropy-production statement is imposed within each regime separately, so nothing here matches the entropy current across the boundary.

The partition is also independent of whether the substrate induces gravity, and the two questions have different answers on the same examples. A film of ^{3}He-A satisfies Sakharov's conditions, while the acoustic condensate of Finazzi–Liberati–Sindoni fails universal coupling [8]: its effective metric is defined only for the phonon sector below the healing length, and the depleted atoms occupying excited single-particle states carry no coupling to that metric. That asymmetry is itself formalized, and a companion paper on emergent gravity through black-hole thermodynamics carries the classification in full [17] (`horizonCrooks_substrate_partition`, `volovikJannes_vs_flsBEC_depletion_asymmetry`).
The cooling branch survives the failure, because a negative $dT_{\rm H}/dt$ constrains the time evolution of the emission temperature on a given background and says nothing about how the dynamics of that background arise. The two questions meet only at the induced-gravity layer, where the Clausius relation $\delta Q = T\,dS$ on every local Rindler horizon returns the Einstein equation as an equation of state [16].

The entropy-production hypothesis is the natural target for replacement by a derivation. Kirklin proves the generalized second law in perturbative gravity to all orders beyond the semiclassical limit and without a UV cutoff, at the cost of a free-energy term for the degrees

of freedom that locate the later horizon cut [15], and Faulkner and Speranza obtain the same law for arbitrary cuts of a Killing horizon from crossed-product algebras [14]. Formalizing either would turn the recorded hypothesis into a consequence. What settles the partition, though, is measurement: a sonic horizon whose emission temperature is observed to fall while it radiates lands in the cooling branch, and the sign of dT_{H}/dt is accessible in a fluid experiment or a lattice simulation long before any end state is reached.

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